TAI LIEU ON TAP: LAP TRINH HUONG DOI TUONG TRONG JAVA

1. Khai niem co ban

Lap trinh huong doi tuong (OOP) la phuong phap lap trinh dua tren khai niem "doi tuong", ma moi doi tuong dong goi du lieu (thuoc tinh) va hanh vi (phuong thuc). Java la nguon ngu huong doi tuong hoan toan, moi chuong trinh deu xay dung quanh class va object.

2. Bon dac trung cua OOP

- Dong goi (Encapsulation): giau thong tin chi tiet ben trong, chi duoc truy cap qua getter/setter. Giup bao ve du lieu va giam phu thuoc giua cac phan.
- Ke thua (Inheritance): lop con ke thua thuoc tinh va phuong thuc cua lop cha bang tu khoa extends. Java chi cho ke thua don (mot lop chi duoc ke thua tu mot lop cha).
- Da hinh (Polymorphism): cung mot phuong thuc nhung hanh vi khac nhau tuy doi tuong. Co 2 loai: override (ghi de) va overload (nap chong).
- Truu tuong (Abstraction): che di chi tiet, chi hien ra tinh nang can thiet qua abstract class hoac interface. Interface chi chua khai bao phuong thuc truoc Java 8.

3. Quan he giua cac thanh phan

Class la ban thiet ke, object la the hien cu the duoc tao tu class bang tu khoa new.
Constructor la phuong thuc dac biet cung ten voi class, duoc goi khi tao object, khong co kieu tra ve. Java co garbage collector tu dong giai phong bo nho.